Unit 4 Binomial Theorem

Intermediate

1. **(a)** Find the first 4 terms in the expansion of $\left(2-\frac{ax}{2}\right)^5$ in ascending powers of x, simplifying each term. [4]

(b) Given that there is no term in x^2 in the expansion of $(2+3x)\left(2-\frac{ax}{2}\right)^5$, find the value of the positive constant a.[2]

(c) Using this value of a, find the coefficient of x^3 in the expansion of $(2+3x)\left(2-\frac{ax}{2}\right)^5$. [2]

[N22/I/5]

2. **(a)** Find the terms in $\frac{1}{x^2}$ and $\frac{1}{x^3}$ in the expansion of $\left(2-\frac{3}{x}\right)^6$. [4]

(b) Given that there is no term in $\frac{1}{x}$ in the expansion of $(x^2+ax)\left(2-\frac{3}{x}\right)^6$, find the value of the constant a. [2]

(c) Using the value of a found in part **(b)**, find the coefficient of x in the expansion of $(x^2+ax)\left(2-\frac{3}{x}\right)^6$. [3]

[N21/II/6]

3. **(i)** By considering the general term in the binomial expansion of $\left(\frac{3}{x^2}+x\right)^8$, explain why every term is dependent on x. [3]

(ii) Find the term independent of x in the expansion of $\left(\frac{3}{x^2}+x\right)^8(5-2x)$. [3]

[N20/II/3]

4. **(i)** Write down and simplify the first three terms in the expansion, in ascending powers of x, of $\left(2-\frac{x}{8}\right)^6$. [3]

(ii) In the expansion of $(4+kx+x^2)\left(2-\frac{x}{8}\right)^6$, the sum of the coefficients of x and x^2 is zero.
Find the value of the constant k. [4]

[N19/I/7]

5. The first three terms in the expansion of $(1-4x)(2+ax)^6$ in ascending powers of x are $64-160x+bx^2$. Find the value of each of the constants a and b. [7]

[N18/II/2]

6. **(i)** By considering the general term in the binomial expansion of $\left(px^3+\frac{1}{x}\right)^9$, where p is a constant, explain why there are no even powers of x in this expansion. [3]

(ii) Given that the coefficient of x^{11} in the expansion of $\left(px^3+\frac{1}{x}\right)^9$ is twice the coefficient of x^7, find the value of p. [4]

[N17/II/3]

7. **(i)** Given that the coefficient of x^2 in the expansion of $(1-2x)^2(1-px)^6$ is 16, find the two possible values of the constant p. [5]

(ii) For each value of p, find the coefficient of x^3 in the expansion of $(1-px)^6$. [3]

[N16/II/2]

8. **(a)** **(i)** Write down, and simplify, the first 4 terms in the expansion of $(1 + x)^9$ in ascending powers of x. [2]

(ii) Replacing x by $z - z^2$, determine the coefficient of z^3 in the expansion of $(1 + z - z^2)^9$. [3]

(b) **(i)** Write down the general term in the binomial expansion of $\left(2x + \dfrac{1}{3x^3}\right)^{10}$. [1]

(ii) Write down the power of x in this general term. [1]

(iii) Hence, or otherwise, determine the coefficient of x^2 in the binomial expansion of $\left(2x + \dfrac{1}{3x^3}\right)^{10}$. [2]

[N15/II/4]